

# 1.2 Memory and storage

Where this unit sits, what it covers, and how far

## Where this unit sits

|                                    |                                                                                                                                                                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qualification                      | GCSE Computer Science (9-1), specification code J277, from 2020                                                                                                                                                                              |
| Component                          | Component 01: Computer systems — 80 marks, 1 hour 30 minutes                                                                                                                                                                                 |
| What OCR says the component covers | the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software                                                                    |
| This unit                          | topic 1.2 Memory and storage                                                                                                                                                                                                                 |
| Checked                            | 3 October 2026, at <a href="https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/">https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/</a> |

**How far this has been checked.** The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

**Before it:** 1.1 Systems architecture

**After it:** 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

## The lesson sequence and the questions it answers

| # | LESSON                               | THE BIG QUESTION                                                                   | ACTIVITIES | MARKS |
|---|--------------------------------------|------------------------------------------------------------------------------------|------------|-------|
| 1 | <b>Memory bay</b><br>ms-101          | What are the differences between RAM and ROM?                                      | 11         | 27    |
| 2 | <b>Swap space</b><br>ms-102          | Why do we need virtual memory, and what can we do to avoid using it?               | 12         | 20    |
| 3 | <b>Storage warehouse</b><br>ms-103   | Why do computers have secondary storage?                                           | 12         | 25    |
| 4 | <b>Top Trumps</b><br>ms-104          | What are the differences between secondary storage devices?                        | 12         | 24    |
| 5 | <b>Pick the storage</b><br>ms-105    | What features of secondary storage make devices suitable for different situations? | 12         | 23    |
| 6 | <b>Units lab</b><br>ms-106           | Why is data stored in binary?                                                      | 15         | 27    |
| 7 | <b>Capacity calculator</b><br>ms-107 | How do you calculate data capacity?                                                | 19         | 27    |

| #  | LESSON                                  | THE BIG QUESTION                                                                                     | ACTIVITIES | MARKS |
|----|-----------------------------------------|------------------------------------------------------------------------------------------------------|------------|-------|
| 8  | <b>Binary workshop</b><br>ms-108        | What can happen when you add two binary numbers together?                                            | 15         | 15    |
| 9  | <b>Assessment 1</b><br>ms-109-test      | assesses the whole unit                                                                              | 0          | 0     |
| 10 | <b>Shift and hex lab</b><br>ms-110      | What is the relationship between denary, binary and hexadecimal?                                     | 15         | 15    |
| 11 | <b>Revision HQ 1.2 part 1</b><br>ms-111 | Which parts of memory and storage can I explain, and which can I only recognise?                     | 21         | 21    |
| 12 | <b>Character foundry</b><br>ms-112      | How does a computer store characters, and what are the implications for the number of bits used?     | 16         | 23    |
| 13 | <b>Pixel studio</b><br>ms-113           | How does a computer store graphics?                                                                  | 15         | 15    |
| 14 | <b>Sound studio</b><br>ms-114           | How do computers store sound, and what are the implications for sample rate, duration and bit depth? | 15         | 15    |
| 15 | <b>Compression works</b><br>ms-115      | What are the effects on a file for each type of compression?                                         | 14         | 24    |
| 16 | <b>Revision HQ 1.2 part 2</b><br>ms-116 | Can I turn each kind of data into numbers, and say what it costs to store?                           | 20         | 27    |
| 17 | <b>Assessment 2</b><br>ms-117-test      | assesses the whole unit                                                                              | 0          | 0     |
| 18 | <b>Binary blitz</b><br>ms-bonus         | stretch work, no new content                                                                         | 1          | 0     |

## Prior knowledge, vocabulary and skills

**Assumed before the unit:** the outcomes of 1.1

**Vocabulary introduced:** RAM, ROM, Virtual memory, Volatile, Secondary storage, Non-volatile, Capacity, Durability, Portability, Suitability, Bit, Byte, Data capacity, File size, Binary, Overflow error, Binary shift, Hexadecimal, Overflow, Character set, Unicode, Pixel, Metadata, Sampling, Sample rate, Lossy compression, Lossless compression, Colour depth, Lossy, Lossless

**What a pupil does, not only reads:** addshift, blitz, calc, code, convert, match, mcq, multi, order, pixels, sort, sound

## Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

**How this was worked out.** The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

| OUTCOME                                              | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE |
|------------------------------------------------------|--------|--------|-----------|----------|----------|
| understand the need for primary storage<br>ms-101-o1 | ms-l01 | yes    | yes       | yes      | Full     |

| OUTCOME                                                                                                   | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE |
|-----------------------------------------------------------------------------------------------------------|--------|--------|-----------|----------|----------|
| know the difference between RAM and ROM<br><b>ms-l01-o2</b>                                               | ms-l01 | yes    | yes       | yes      | Full     |
| know the purpose of RAM and ROM<br><b>ms-l01-o3</b>                                                       | ms-l01 | yes    | yes       | yes      | Full     |
| understand the need for virtual memory<br><b>ms-l02-o1</b>                                                | ms-l02 | yes    | yes       | yes      | Full     |
| understand how virtual memory works<br><b>ms-l02-o2</b>                                                   | ms-l02 | yes    | yes       | yes      | Full     |
| understand the need for secondary storage<br><b>ms-l03-o1</b>                                             | ms-l03 | yes    | yes       | yes      | Full     |
| know the common types of storage<br><b>ms-l03-o2</b>                                                      | ms-l03 | yes    | yes       | yes      | Full     |
| know how each type works<br><b>ms-l03-o3</b>                                                              | ms-l03 | yes    | yes       | yes      | Full     |
| know the characteristics of storage devices<br><b>ms-l04-o1</b>                                           | ms-l04 | yes    | yes       | yes      | Full     |
| be able to compare devices using those characteristics<br><b>ms-l04-o2</b>                                | ms-l04 | yes    | yes       | yes      | Full     |
| understand the suitability of storage devices for certain applications<br><b>ms-l05-o1</b>                | ms-l05 | yes    | yes       | yes      | Full     |
| understand the advantages and disadvantages of devices based on their characteristics<br><b>ms-l05-o2</b> | ms-l05 | yes    | yes       | yes      | Full     |
| know the units bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte and petabyte<br><b>ms-l06-o1</b> | ms-l06 | yes    | yes       | yes      | Full     |
| be able to convert between units<br><b>ms-l06-o2</b>                                                      | ms-l06 | yes    | yes       | yes      | Full     |
| understand that data must be converted to binary to be processed<br><b>ms-l06-o3</b>                      | ms-l06 | yes    | yes       | yes      | Full     |
| know what data capacity means<br><b>ms-l07-o1</b>                                                         | ms-l07 | yes    | yes       | yes      | Full     |
| understand how to calculate data capacity requirements<br><b>ms-l07-o2</b>                                | ms-l07 | yes    | yes       | yes      | Full     |
| convert positive denary whole numbers (0-255) to 8-bit binary and back<br><b>ms-l08-o1</b>                | ms-l08 | yes    | yes       | yes      | Full     |
| add two 8-bit binary integers<br><b>ms-l08-o2</b>                                                         | ms-l08 | yes    | yes       | yes      | Full     |
| understand the terms most and least significant bit<br><b>ms-l08-o3</b>                                   | ms-l08 | yes    | yes       | yes      | Full     |

| OUTCOME                                                                                                          | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE    |
|------------------------------------------------------------------------------------------------------------------|--------|--------|-----------|----------|-------------|
| understand how overflow errors occur<br><a href="#">ms-l08-o4</a>                                                | ms-l08 | yes    | yes       | yes      | Full        |
| know how to perform left and right binary shifts<br><a href="#">ms-l10-o1</a>                                    | ms-l10 | yes    | yes       | yes      | Full        |
| understand what a binary shift achieves<br><a href="#">ms-l10-o2</a>                                             | ms-l10 | yes    | yes       | yes      | Full        |
| convert denary to 2-digit hexadecimal and back<br><a href="#">ms-l10-o3</a>                                      | ms-l10 | yes    | yes       | yes      | Full        |
| convert between binary and hexadecimal<br><a href="#">ms-l10-o4</a>                                              | ms-l10 | yes    | yes       | yes      | Full        |
| know which parts of the topic I am secure on<br><a href="#">ms-l11-o1</a>                                        | ms-l11 | —      | —         | —        | Not covered |
| finish any outstanding tasks from lessons 1 to 10<br><a href="#">ms-l11-o2</a>                                   | ms-l11 | —      | —         | —        | Not covered |
| understand that all data, including text, is represented in binary<br><a href="#">ms-l12-o1</a>                  | ms-l12 | yes    | yes       | yes      | Full        |
| know what is meant by a character set<br><a href="#">ms-l12-o2</a>                                               | ms-l12 | yes    | yes       | yes      | Full        |
| understand the relationship between the number of bits and the number of characters<br><a href="#">ms-l12-o3</a> | ms-l12 | yes    | yes       | yes      | Full        |
| know about ASCII and Unicode<br><a href="#">ms-l12-o4</a>                                                        | ms-l12 | yes    | yes       | yes      | Full        |
| understand how an image is a series of pixels represented in binary<br><a href="#">ms-l13-o1</a>                 | ms-l13 | yes    | yes       | yes      | Full        |
| know what is meant by metadata<br><a href="#">ms-l13-o2</a>                                                      | ms-l13 | yes    | yes       | yes      | Full        |
| understand the effect of colour depth and resolution on file size<br><a href="#">ms-l13-o3</a>                   | ms-l13 | yes    | yes       | yes      | Full        |
| understand how sound is sampled and stored in digital form<br><a href="#">ms-l14-o1</a>                          | ms-l14 | yes    | yes       | yes      | Full        |
| understand how sample rate, duration and bit depth affect file size and quality<br><a href="#">ms-l14-o2</a>     | ms-l14 | yes    | yes       | yes      | Full        |
| know why data is compressed<br><a href="#">ms-l15-o1</a>                                                         | ms-l15 | yes    | yes       | yes      | Full        |
| understand the difference between lossy and lossless compression<br><a href="#">ms-l15-o2</a>                    | ms-l15 | yes    | yes       | yes      | Full        |
| know why some data can only be compressed losslessly<br><a href="#">ms-l15-o3</a>                                | ms-l15 | yes    | yes       | yes      | Full        |

| OUTCOME                                                                          | LESSON | TAUGHT | PRACTISED | ASSESSED | COVERAGE    |
|----------------------------------------------------------------------------------|--------|--------|-----------|----------|-------------|
| know which parts of data representation I am secure on<br><span>ms-l16-o1</span> | ms-l16 | yes    | yes       | yes      | Full        |
| finish outstanding tasks from lessons 12 to 15<br><span>ms-l16-o2</span>         | ms-l16 | —      | —         | —        | Not covered |

38 full · 0 partial · 3 not covered, of 41 outcomes.

## Gaps, assumptions and overlap

None found by the trace.

### What could not be traced

The wall text of ms - bonus is painted into the 360 image and held nowhere as text, so an outcome in that lesson lesson can only be traced through its activities and its exam questions. Read the rows above for it with that in mind.

Revise 360 — a product of Revise 360 Ltd, created by Olly Pettitt. An independent resource: not written, endorsed or approved by OCR or any exam board. Classroom use and personal study only.